

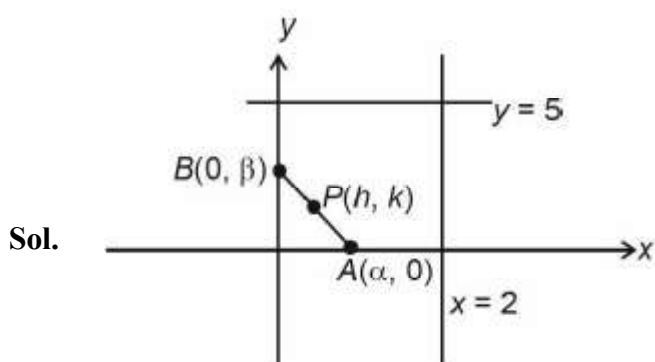
41. Let R be a rectangle given by the lines $x = 0$, $x = 2$, $y = 0$ and $y = 5$. Let $A(\alpha, 0)$ and $B(0, \beta)$, $\alpha \in [0, 2]$ and $\beta \in [0, 5]$, be such that the line segment AB divides the area of the rectangle R in the ratio 4 : 1. Then, the mid-point of AB lies on a :

(A) circle (B) straight line (C) hyperbola (D) parabola

माना रेखाओं $x = 0$, $x = 2$, $y = 0$ तथा $y = 5$ द्वारा निर्मित आयत R है। माना $A(\alpha, 0)$ तथा $B(0, \beta)$, $\alpha \in [0, 2]$, $\beta \in [0, 5]$, इस प्रकार हैं कि रेखाखंड AB आयत R क्षेत्रफल को 4 : 1 के अनुपात में बाँटती है। तो AB का मध्य बिंदु एक

(A) वृत्त (B) सरल रेखा (C) अतिपरवलय (D) परवलय

Ans. C



$$\frac{10 - \frac{1}{2}\alpha\beta}{\frac{1}{2}\alpha\beta} = \frac{4}{1} \Rightarrow 20 - \alpha\beta = 4\alpha\beta$$

$$\Rightarrow \alpha\beta = 4$$

$$\text{Let } h = \frac{\alpha}{2}, \beta = \frac{k}{2}$$

$$\therefore 4hk = 4$$

$$\therefore xy = 1$$

42. Let the centroid of an equilateral triangle ABC be at the origin. Let one of the sides of the equilateral triangle be along the straight line $x + y = 3$. If R and r be the radius of circumcircle and incircle respectively of $\triangle ABC$, then $(R + r)$ is equal to :

माना एक समबाहु त्रिभुज ABC का केन्द्रक मूलबिन्दु पर है। माना इस त्रिभुज की एक भुजा सरल रेखा $x + y = 3$ के अनुदिश है। यदि $\triangle ABC$ के परिवृत्त तथा अंतवृत्त की त्रिज्याएँ क्रमशः R तथा r हैं, तो $(R + r)$ बराबर है :

- (A) $3\sqrt{2}$ (B) $\frac{9}{\sqrt{2}}$ (C) $2\sqrt{2}$ (D) $7\sqrt{2}$

Ans. B

Sol. $r = \frac{\left| \frac{-3}{\sqrt{2}} \right|}{\frac{1}{\sqrt{2}}} = \frac{3}{\sqrt{2}}$

$$r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$\frac{3}{\sqrt{2}} = 4R \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

$$R = 3\sqrt{2}$$

$$R + r = \frac{3}{\sqrt{2}} + 3\sqrt{2} = \frac{9}{\sqrt{2}}$$

43. A circle cuts a chord of length $4a$ on the x -axis and passes through a point on the y -axis, distant $2b$ from the origin. Then the locus of the centre of this circle, is :

- (A) A hyperbola (B) A parabola (C) A straight line (D) An ellipse

एक वृत्त x -अक्ष पर एक जीवा काटता है जिसकी लंबाई $4a$ है तथा यह वृत्त y -अक्ष के एक बिन्दु से हो कर जाता है जिसकी मूलबिन्दु से दूरी $2b$ है। तो वृत्त के केन्द्र का बिन्दुपथ है –

- (A) एक अतिपरवलय (B) एक परवलय (C) एक सरल रेखा (D) एक दीर्घवृत्त

Ans. B

Sol. $x^2 + y^2 + 2gx + 2fy + c = 0$

$$0 + 4b^2 + 4bf + c = 0$$

$$4b^2 + 4bf + c = 0$$

$$2\sqrt{g^2 - c} = 4a$$

$$g^2 - c = 4a^2$$

$$g^2 - 4a^2 = c$$

$$4b^2 + 4bf + g^2 - 4a^2 = 0$$

$$x^2 + 4by + 4b^2 - 4a^2 = 0$$

parabola

44. It a line, $y = mx + c$ is a tangent to the circle, $(x - 3)^2 + y^2 = 1$ and it is perpendicular to a line L_1 , where L_1

is the tangent to the circle, $x^2 + y^2 = 1$ at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$; then :

यदि एक रेखा $y = mx + c$, वृत्त $(x - 3)^2 + y^2 = 1$ की एक स्पर्श रेखा है तथा यह एक रेखा L_1 पर लम्ब है, जहाँ L_1 वृत्त $x^2 + y^2 =$

1 के बिन्दु $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ पर स्पर्श रेखा है, तो :

- (A) $c^2 + 7c + 6 = 0$ (B) $c^2 - 7c + 6 = 0$ (C) $c^2 + 6c + 7 = 0$ (D) $c^2 - 6c + 7 = 0$

Ans. C

Sol. Equation $L_1 \Rightarrow x \cdot \frac{1}{\sqrt{2}} + y \cdot \frac{1}{\sqrt{2}} = 1$

$$x + y = \sqrt{2}$$

$$m_{L_1} = -1$$

$$m = 1$$

$$C : (x - 3)^2 + y^2 = 1$$

$$L : y = x + c \quad \Rightarrow x - y + c = 0$$

$$\text{Condition of tangency for circle} = \left| \frac{3 - 0 + c}{\sqrt{2}} \right| = 1$$

$$(c + 3)^2 = 2$$

$$c^2 + 6c + 7 = 0$$

45. If the point $\left(\alpha, \frac{7\sqrt{3}}{3}\right)$ lies on the curve traced by the mid-points of the line segments of the lines

$x \cos \theta + y \sin \theta = 7, \theta \in \left(0, \frac{\pi}{2}\right)$ between the co-ordinates axes, then α is equal to :

यदि रेखाओं $x \cos \theta + y \sin \theta = 7, \theta \in \left(0, \frac{\pi}{2}\right)$ के निर्देशांक अक्षों के बीच रेखाखंडों के मध्य बिंदुओं द्वारा बने वक्र पर एक बिंदु

$\left(\alpha, \frac{7\sqrt{3}}{3}\right)$ है, तो α बराबर है :

- (A) $7\sqrt{3}$ (B) $-7\sqrt{3}$ (C) -7 (D) 7

Ans. D

Sol. $x \cos \theta + y \sin \theta = 7; \theta \in \left(0, \frac{\pi}{2}\right)$

Mid point of line segment between coordinate axes

$$(h, k) \equiv \left(\frac{7}{2 \cos \theta}, \frac{7}{2 \sin \theta}\right)$$

$$\Rightarrow \text{Locus of } (h, k) \text{ is } \frac{49}{y^2} + \frac{49}{x^2} = 4 \Rightarrow \frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{49}$$

Point $\left(\alpha, \frac{7\sqrt{3}}{3}\right)$ lies on it

$$\Rightarrow \frac{1}{\alpha^2} + \frac{3}{49} = \frac{4}{49}$$

$$\Rightarrow \frac{1}{\alpha^2} = \frac{1}{49}$$

$$\Rightarrow \alpha = 7$$

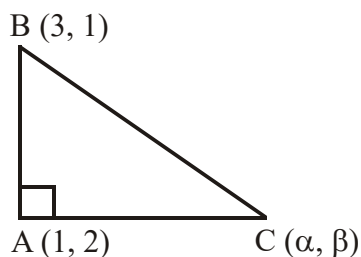
46. A triangle ABC lying in the first quadrant has two vertices as A(1, 2) and B(3, 1). If $\angle BAC = 90^\circ$, and $\text{ar}(\triangle ABC) = 5\sqrt{5}$ sq. units, then the abscissa of the vertex C is :

एक त्रिभुज ABC जो प्रथम चतुर्थांश में स्थित है, के दो शीर्ष बिन्दु A(1, 2) तथा B(3, 1) हैं। यदि $\angle BAC = 90^\circ$ तथा $\text{ar}(\triangle ABC) = 5\sqrt{5}$ वर्ग इकाई हैं, तो शीर्ष बिन्दु C का भुज है :

- (A) $1 + \sqrt{5}$ (B) $2 + \sqrt{5}$ (C) $2\sqrt{5} - 1$ (D) $1 + 2\sqrt{5}$

Ans. D

Sol.



$$m_{AB} \times m_{AC} = -1$$

$$\Rightarrow \frac{\beta - 2}{\alpha - 1} \cdot \frac{2 - 1}{1 - 3} = -1$$

$$\Rightarrow \frac{\beta - 2}{\alpha - 1} \cdot \left(-\frac{1}{2}\right) = -1$$

$$\Rightarrow \frac{\beta - 2}{\alpha - 1} = 2$$

$$\Rightarrow \beta - 2 = 2\alpha - 2$$

$$\beta = 2\alpha \dots\dots\dots (1)$$

$$\text{Area of } \triangle ABC = 5\sqrt{5}$$

$$\frac{1}{2} \times AB \times AC = 5\sqrt{5}$$

$$\Rightarrow AB^2 \times AC^2 = 125 \times 4$$

$$\left((\alpha-1)^2 + (\beta-2)^2 \right) \left((1-3)^2 + (2-1)^2 \right) = 500$$

$$\left((\alpha-1)^2 + 4(\alpha-1)^2 \right) (4+1) = 500$$

$$\Rightarrow 5(\alpha-1)^2 \cdot 5 = 500$$

$$(\alpha-1)^2 = 20 \quad |\alpha-1| = 2\sqrt{5} \quad \therefore \alpha = 1 \pm 2\sqrt{5}$$

$$\therefore \alpha = 1 + 2\sqrt{5}, \text{ As } \triangle ABC \text{ lying in first quadrant.}$$

47. The intersection of three lines $x - y = 0$, $x + 2y = 3$ and $2x + y = 6$ is a :

(A) Isosceles triangle

(B) Right angled triangle

(C) None of the above

(D) Equilateral triangle

तीन रेखाओं $x - y = 0$, $x + 2y = 3$ तथा $2x + y = 6$ का प्रतिच्छेदन :

(A) एक समद्विबाहु त्रिभुज है

(B) एक समकोण त्रिभुज है

(C) इनमें से कोई नहीं है

(D) एक समबाहु त्रिभुज है

Ans. A

Sol. Slopes of the 3 given lines in order are

$$m_1 = 1, m_2 = -\frac{1}{2}, m_3 = -2$$

So the internal angles of the triangle will be

$$\tan A = \frac{m_1 - m_2}{1 + m_1 m_2} = 3$$

$$\tan C = \frac{m_3 - m_1}{1 + m_3 m_1} = 3$$

$$\tan B = \frac{m_2 - m_3}{1 + m_2 m_3} = -3$$

So, it is an Isosceles Triangle.

48. If the area of the triangle formed by the positive x-axis, the normal and the tangent to the circle $(x-2)^2 + (y-3)^2 = 25$ at the point $(5, 7)$ is A, then $24A$ is equal to _____.

यदि त्रिभुज, जो धनात्मक x-अक्ष तथा वृत्त $(x-2)^2 + (y-3)^2 = 25$ के बिन्दु $(5, 7)$ पर खींचे गए अभिलम्ब तथा स्पर्श रेखा द्वारा बनता है, का क्षेत्रफल A है, तो $24A$ बराबर है _____ ।

Ans. 1225

Sol. Normal at P $(5, 7)$

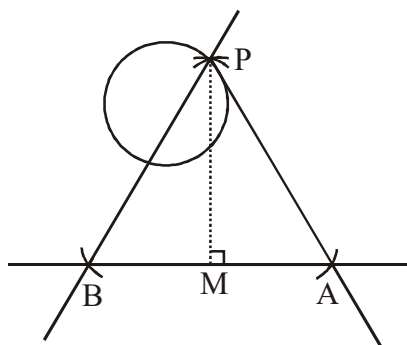
$$4x - 3y + 1 = 0$$

$$\Rightarrow B\left(-\frac{1}{4}, 0\right)$$

Tangent at P

$$3x + 4y = 43$$

$$\Rightarrow A\left(\frac{43}{3}, 0\right)$$



PB is normal at P, PA is tangent at P

$$\text{Area} = \frac{1}{2} (AB) (PM)$$

$$A = \frac{1}{2} \left(\frac{43}{3} + \frac{1}{4} \right) (7) = \frac{1225}{24}$$

$$24A = 1225$$

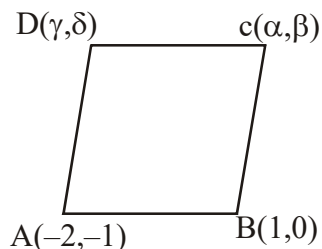
49. Let $A(-2, -1)$, $B(1, 0)$, $C(\alpha, \beta)$ and $D(\gamma, \delta)$ be the vertices of a parallelogram ABCD. If the point C lies on

$2x - y = 5$ and the point D lies on $3x - 2y = 6$, then the value of $|\alpha + \beta + \gamma + \delta|$ is equal to _____.

माना एक समांतर चतुर्भुज ABCD के शीर्ष $A(-2, -1)$, $B(1, 0)$, $C(\alpha, \beta)$ तथा $D(\gamma, \delta)$ हैं। यदि बिन्दु C, रेखा $2x - y = 5$ पर है तथा बिन्दु D, रेखा $3x - 2y = 6$ पर है, तो $|\alpha + \beta + \gamma + \delta|$ का मान बराबर है _____

Ans. 32

Sol.



$$2\alpha - \beta = 5 \dots\dots\dots(i)$$

$$3\gamma - 2\delta = 6 \dots\dots\dots(ii)$$

Diagonal bisect each other

$$\alpha - 2 = \gamma + 1 \Rightarrow \alpha = \gamma + 3$$

$$\beta - 1 = \delta \Rightarrow \beta = \delta + 1$$

Now form (i)

$$2(\gamma + 3) - \delta - 1 = 5 \Rightarrow \delta = 2\gamma$$

$$\text{From (ii) } \gamma = -6, \delta = -12, \alpha = -3, \beta = -11$$

$$|\alpha + \beta + \gamma + \delta| = 32$$

50. If the sum of squares of all real values of α , for which the lines $2x - y + 3 = 0$, $6x + 3y + 1 = 0$ and $\alpha x + 2y - 2 = 0$ do not form a triangle is p, then the greatest integer less than or equal to p is _____.

माना α के सभी वास्तविक मानों, जिनके लिए रेखाएँ $2x - y + 3 = 0$, $6x + 3y + 1 = 0$ तथा $\alpha x + 2y - 2 = 0$ एक त्रिभुज नहीं बनाती हैं, के वर्गों का योग p है, तो महत्तम पूर्णांक $\leq p$ है _____

Ans. 32

Sol. $2x - y + 3 = 0$

$$6x + 3y + 1 = 0$$

$$\alpha x + 2y - 2 = 0$$

will not form a Δ if $\alpha x + 2y - 2 = 0$ is concurrent with $2x - y + 3 = 0$ and $6x + 3y + 1 = 0$ or parallel to either of them so

Case-1: Concurrent lines

$$\begin{vmatrix} 2 & -1 & 3 \\ 6 & 3 & 1 \\ \alpha & 2 & -2 \end{vmatrix} = 0 \Rightarrow \alpha = \frac{4}{5}$$

Case-2 : Parallel lines

$$-\frac{\alpha}{2} = \frac{-6}{3} \text{ or } -\frac{\alpha}{2} = 2$$

$$\Rightarrow \alpha = 4 \text{ or } \Rightarrow \alpha = -4$$

$$P = 16 + 16 + \frac{16}{25}$$

$$[P] = \left[32 + \frac{16}{25} \right] = 32$$

51. Let a common tangent to the curves $y^2 = 4x$ and $(x - 4)^2 + y^2 = 16$ touch the curves at the points P and Q. Then $(PQ)^2$ is equal to _____.

माना वक्रों $y^2 = 4x$ तथा $(x - 4)^2 + y^2 = 16$ की उभयनिष्ठ स्पर्श रेखा, वक्रों को बिंदुओं P तथा Q पर स्पर्श करती है। तो $(PQ)^2$ बराबर है _____

Ans. 32

Sol. $y = mx + \frac{1}{m}$

$$\perp r \text{ from } (4, 0) = 4$$

$$\left| \frac{4m + \frac{1}{m}}{m^2 + 1} \right| = 4$$

$$\text{Or } 16m^2 + \frac{1}{m^2} + 8 = 16m^2 + 16$$

$$\text{Or } m^2 = \frac{1}{8}, \text{ if } m = \frac{1}{2\sqrt{2}} \text{ then}$$

$$P = \left(\frac{a}{m^2}, \frac{2a}{m} \right) = (8, 4\sqrt{2})$$

Q = foot of $\perp r$ from (4, 0) on tangent

$$= \left(\frac{8}{3}, \frac{8\sqrt{2}}{3} \right)$$

$$(PQ)^2 = \left(\frac{16}{3} \right)^2 + \left(\frac{4\sqrt{2}}{3} \right)^2$$

$$= \frac{256+32}{9} = 32$$

52. Let m_1 and m_2 be the slopes of the tangents drawn from the point $P(4, 1)$ to the hyperbola $H: \frac{y^2}{25} - \frac{x^2}{16} = 1$. If Q is the point from which the tangents drawn to H have slopes $|m_1|$ and $|m_2|$ and they make positive intercepts α and β on the x -axis, then $\frac{(PQ)^2}{\alpha\beta}$ is equal to _____.

माना बिन्दु $P(4, 1)$ से अतिपरवलय $H: \frac{y^2}{25} - \frac{x^2}{16} = 1$ पर खींची गई स्पर्श रेखाओं की प्रवणताएं m_1 तथा m_2 हैं। यदि Q वह बिन्दु है, जिससे H पर खींची गई स्पर्श रेखाओं की प्रवणताएं $|m_1|$ तथा $|m_2|$ हैं तथा यह स्पर्श रेखाएं x -अक्ष पर धनात्मक अंतःखण्ड α तथा β बनाती है, तो $\frac{(PQ)^2}{\alpha\beta}$ बराबर है _____

Ans. 8

Sol. Equation of tangent to the hyperbola $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$

$$y = mx \pm \sqrt{a^2 - b^2 m^2}$$

passing through $(4, 1)$

$$1 = 4m \pm \sqrt{25 - 16m^2} \Rightarrow 4m^2 - m - 3 = 0$$

$$\Rightarrow m = 1, \frac{-3}{4}$$

Equation of tangent with positive slopes 1 & $\frac{3}{4}$.

$$\left. \begin{array}{l} 4y = 3x - 16 \\ y = x - 3 \end{array} \right\} \text{ with positive intercept on } x\text{-axis}$$

$$\alpha = \frac{16}{3}, \beta = 3$$

Intersection points :

$$Q: (-4, -7)$$

$$P: (4, 1)$$

$$PQ^2 = 128$$

$$\frac{PQ^2}{\alpha\beta} = \frac{128}{16} = 8$$

53. A circle with centre (2,3) and radius 4 intersects the line $x + y = 3$ at the points P and Q . If the tangents at P and Q intersect at the point $S (\alpha, \beta)$, then $4\alpha - 7\beta$ is equal to _____.

एक वृत्त, जिसका केन्द्र (2,3) है तथा त्रिज्या 4 है, रेखा $x + y = 3$ को बिंदुओं P तथा Q पर काटता है। यदि P तथा Q पर स्पर्श रेखाएँ बिंदु $S (\alpha, \beta)$ पर मिलती हैं, तो $4\alpha - 7\beta$ बराबर है।

Ans. 11

Sol. The given line is polar of $P(2, \beta)$ w.r.t. given circle

$$x^2 + y^2 - 4x - 6y - 3 = 0$$

Chord or contact

$$\alpha x + \beta y - 2(x + \alpha) - 3(y + \beta) - 3 = 0$$

$$\Rightarrow (\alpha - 2)x + (\beta - 3)y - (2\alpha + 3\beta + 3) = 0 \quad \dots(i)$$

$\therefore$ But the equation of chord of contact is given

$$\text{as : } x + y - 3 = 0 \quad \dots(ii)$$

comparing the coefficients

$$\frac{\alpha - 2}{1} = \frac{\beta - 3}{1} = -\left(\frac{2\alpha + 3\beta + 3}{-3}\right)$$

$$\text{On solving} \quad \alpha = -6$$

$$\beta = -5$$

$$\text{Now} \quad 4\alpha - 7\beta = 11$$

54. The line $x = 8$ is the directrix of the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with the corresponding focus (2,0). If the tangent to E at the point P in the first quadrant passes through the point $(0, 4\sqrt{3})$ and intersects the x -axis at Q , then $(3PQ)^2$ is equal to _____.

दीर्घवृत्त $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ की नियता $x = 8$ है तथा संगत नाभि (2,0) है। यदि प्रथम चतुर्थांश में E के बिंदु P पर स्पर्श रेखा, बिंदु $(0, 4\sqrt{3})$ से होकर जाती है तथा x -अक्ष को Q पर काटती है, तो $(3PQ)^2$ बराबर है।

Ans. 39

Sol. $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$ae = 2$$

$$\frac{a}{e} = 8$$

$$e = \frac{1}{2}$$

$$a^2 = 16$$

$$a = 4$$

$$b^2 = a^2(1 - e^2) = 16\left(1 - \frac{1}{4}\right) = 16 \times \frac{3}{4} = 12$$

$$\therefore \text{Ellipse is } \frac{x^2}{16} + \frac{y^2}{12} = 1$$

$$\text{Tangent at P is } \frac{x \cos \theta}{4} + \frac{y \sin \theta}{2\sqrt{3}} = 1$$

$$\text{Passing through } (0, 4\sqrt{3})$$

$$\therefore 0 + \frac{4\sqrt{3} \sin \theta}{2\sqrt{3}} = 1$$

$$\sin \theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

$$\therefore \text{Tangent is } \frac{x}{4} \cdot \frac{\sqrt{13}}{2} + \frac{y}{2\sqrt{3}} \cdot \frac{1}{2} = 1$$

$$\frac{\sqrt{3}x}{8} + \frac{y}{4\sqrt{3}} = 1$$

$$Q\left(\frac{8}{\sqrt{3}}, 0\right) \quad P(4 \cos \theta, 2\sqrt{3} \sin \theta)$$

$$P(2\sqrt{3}, \sqrt{3})$$

$$PQ = \sqrt{\left(\frac{8}{\sqrt{3}} - 2\sqrt{3}\right)^2 + 3} = \sqrt{\left(\frac{2}{\sqrt{3}}\right)^2 + 3} = \sqrt{\frac{4}{3} + 3}$$

$$= \sqrt{\frac{13}{3}}$$

$$PQ^2 = \frac{13}{3} \Rightarrow (3PQ)^2 = \frac{13}{3} \times 9 = 39$$

55. Let for any three distinct consecutive terms a, b, c of an A.P., the lines $ax + by + c = 0$ be concurrent at the point P and $Q(\alpha, \beta)$ be a point such that the system of equations

$$x + y + z = 6,$$

$$2x + 5y + \alpha z = \beta \text{ and}$$

$$x + 2y + 3z = 4, \text{ has infinitely many solutions. Then } (PQ)^2 \text{ is equal to } \underline{\hspace{2cm}}.$$

माना एक A.P. के किसी भी तीन भिन्न क्रमागत पदों a, b, c के लिए रेखाएँ $ax + by + c = 0$ एक बिन्दु P पर संगामी हैं तथा बिन्दु $Q(\alpha, \beta)$ के लिए समीकरण निकाय

$$x + y + z = 6,$$

$$2x + 5y + \alpha z = \beta \text{ तथा}$$

$$x + 2y + 3z = 4, \text{ के अनंत हल है। तो } (PQ)^2 \text{ बराबर है } \underline{\hspace{2cm}}.$$

Ans. 113

Sol. $b - a = c - b$

$$a - 2b + c = 0$$

$$\Rightarrow ax + by + c = 0 \text{ are concurrent at } P(1, -2)$$

$$\therefore x + y + z = 6,$$

$$2x + 5y + \alpha z = \beta,$$

$$x + 2y + 3z = 4$$

$$\Delta = 8 - \alpha,$$

$$\Delta_1 = -8\alpha - \beta + 70,$$

$$\Delta_2 = 2\alpha + 2\beta - 28,$$

$$\Delta_3 = 6 - \beta \text{ for infinitely many solutions}$$

$$\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$$

$$\Rightarrow \alpha = 8, \beta = 6$$

$$\Rightarrow Q(8, 6), P(1, -2)$$

$$\Rightarrow (PQ)^2 = 113$$

56. Let the normal at a point P on the curve $y^2 - 3x^2 + y + 10 = 0$ intersect the y -axis at $\left(0, \frac{3}{2}\right)$. If m is the slope

of the tangent at P to the curve, then $|m|$ is equal to _____.

माना वक्र $y^2 - 3x^2 + y + 10 = 0$ के बिन्दु P पर खींचा गया अभिलम्ब y-अक्ष को $\left(0, \frac{3}{2}\right)$ पर काटता है। यदि P पर वक्र की स्पर्श रेखा का ढाल m है, तो $|m|$ बराबर है _____।

Ans. 4

Sol. $p(x_1, y_1)$

$$2y \frac{dy}{dx} - 6x + \frac{dy}{dx} = 0$$

$$y_1 = 1$$

$$\frac{dy}{dx} = \frac{6x_1}{1 + 2y_1}$$

$$x_1 \pm 2$$

$$\frac{\frac{3}{2} - y_1}{-x_1} = -\frac{1 + 2y_1}{6x_1}$$

$$\text{slope of tangent} = \pm 4$$

57. Let ABC be an isosceles triangle in which A is at $(-1, 0)$, $\angle A = \frac{2\pi}{3}$, $AB = AC$ and B is on the positive x-axis.

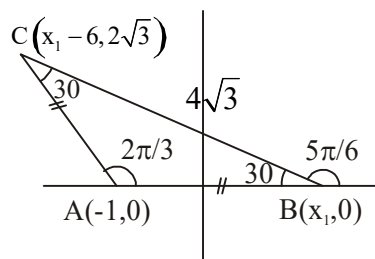
If $BC = 4\sqrt{3}$ and the line BC intersects the line $y = x + 3$ at (α, β) , then $\frac{\beta^4}{\alpha^2}$ is _____.

माना एक समद्विबाहु त्रिभुज ABC में A, बिंदु $(-1, 0)$ हैं, $\angle A = \frac{2\pi}{3}$ हैं, $AB = AC$ है तथा B, धनात्मक x-अक्ष पर है। यदि

$BC = 4\sqrt{3}$ तथा रेखा BC, रेखा $y = x + 3$ को (α, β) पर काटती है, तो $\frac{\beta^4}{\alpha^2}$ बराबर है _____

Ans. 36

Sol.



Co-ordinates of 'C' are

$$\frac{x - x_1}{\cos \frac{5\pi}{6}} = \frac{y - 0}{\sin \frac{5\pi}{6}} = 4\sqrt{3}$$

$$x = 4\sqrt{3} \cos \frac{5\pi}{6} + x_1 \Rightarrow 4\sqrt{3} \times \left(-\frac{\sqrt{3}}{2}\right) + x_1 \Rightarrow (x_1 - 6)$$

$$y = 4\sqrt{3} \sin \frac{5\pi}{6} \Rightarrow 4\sqrt{3} \times \frac{1}{2} \Rightarrow 2\sqrt{3}$$

$$C(x_1 - 6, 2\sqrt{3})$$

$$m_{AC} = -\sqrt{3}$$

$$\frac{0 - 2\sqrt{3}}{-1 - x_1 + 6} = -\sqrt{3} \Rightarrow x_1 = 3$$

$$BC : x + \sqrt{3}y = 3, y = x + 3$$

$$y - 3 + \sqrt{3}y = 3$$

$$y = \frac{6}{\sqrt{3} + 1}, x = \frac{6}{\sqrt{3} + 1} - 3 \Rightarrow \frac{6 - 3\sqrt{3} - 3}{\sqrt{3} + 1}$$

$$x = \frac{3 - 3\sqrt{3}}{\sqrt{3} + 1}$$

$$\frac{-6}{(1 + \sqrt{3})^2}$$

$$\text{So, } \frac{\beta^4}{\alpha^2} = 36$$

58. Consider the triangle with vertices A(2, 1), B(0, 0) and C(t, 4), $t \in [0, 4]$. If the maximum and the minimum perimeters of such triangles are obtained at $t = \alpha$ and $t = \beta$ respectively, then $6\alpha + 21\beta$ is equal to _____.

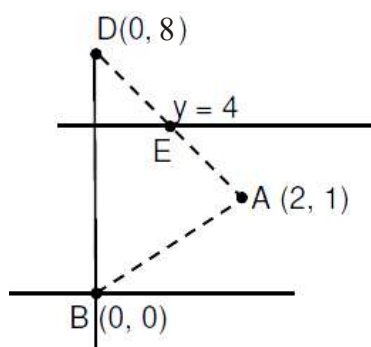
एक त्रिभुज इस प्रकार है जिसके शीर्ष A(2, 1), B(0, 0) तथा C(t, 4), $t \in [0, 4]$ है। इस प्रकार के त्रिभुजों का अधिकतम तथा न्यूनतम परिमाप क्रमशः $t = \alpha$ तथा $t = \beta$ पर प्राप्त होता है तो $6\alpha + 21\beta$ का मान होगा _____

Ans. 48

Sol. A (2, 1), B(0, 0), C (t, 4), $t \in [0, 4]$

Image of B in $y = 4$

D(0, 8)



Equation of AD :

$$y - 1 = \frac{7}{-2}(x - 2)$$

$$7x + 2y = 16$$

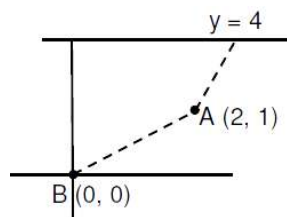
$$E\left(\frac{8}{7}, 4\right) \Rightarrow \beta = \frac{8}{7}$$

For maximum perimeter

Maximum perimeter at C (4, 4)

$$\therefore t \in [0, 4]$$

$$\text{Now, } 6\alpha + 21\beta = 6 \times 4 + 21\left(\frac{8}{7}\right) = 48$$

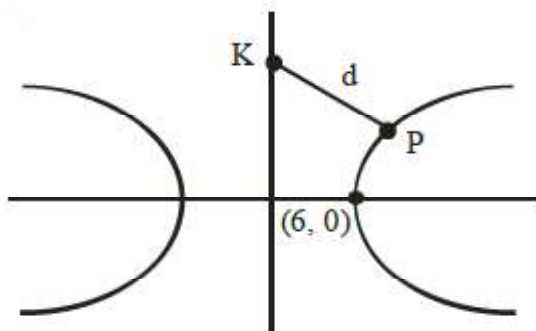


59. The vertices of hyperbola H are $(\pm 6, 0)$ and its eccentricity is $\frac{\sqrt{5}}{2}$. Let N be the normal to H at a point in the first quadrant and parallel to the line $\sqrt{2}x + y = 2\sqrt{2}$. If d is the length of the line segment of N between H and the y-axis then d^2 is equal to _____.

एक अतिपरवलय H के शीर्ष $(\pm 6, 0)$ है तथा उत्केन्द्रता $\frac{\sqrt{5}}{2}$ है। माना प्रथम चतुर्थांश में H के एक बिंदु पर रेखा $\sqrt{2}x + y = 2\sqrt{2}$ के समांतर अभिलंब N है। यदि N के H तथा y-अक्ष के बीच रेखाखंड की लंबाई d है, तो d^2 बराबर है।

Ans. 216

Sol.



$$H: \frac{x^2}{36} - \frac{y^2}{9} = 1$$

equation of normal is $6x \cos\theta + 3y \cot\theta = 45$

$$\text{slope} = -2 \sin\theta = -\sqrt{2}$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

Equation of normal is $\sqrt{2}x + y = 15$

$$P: (a \sec\theta, b \tan\theta)$$

$$\Rightarrow P(6\sqrt{2}, 3) \text{ and } K(0, 15)$$

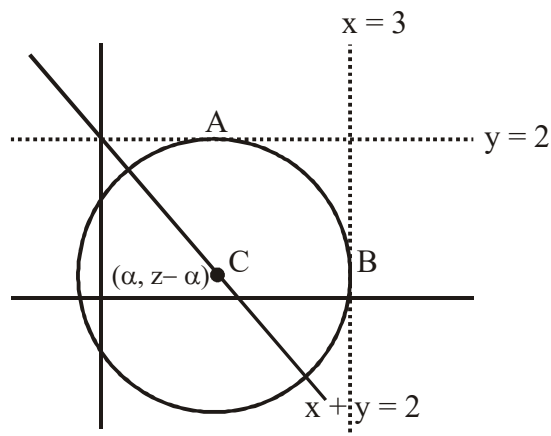
$$d^2 = 216$$

60. The diameter of the circle, whose centre lies on the line $x + y = 2$ in the first quadrant and which touches both the lines $x = 3$ and $y = 2$, is _____.

वृत्त, जिसका केन्द्र प्रथम चतुर्थांश में रेखा $x + y = 2$ पर है तथा जो दोनों रेखाओं $x = 3$ तथा $y = 2$ को स्पर्श करता है, का व्यास है _____।

Ans. 3

Sol.



$$CA = CB$$

$$|2 - (2 - \alpha)| = |3 - \alpha|$$

$$|\alpha| = |\alpha - 3|$$

$$\alpha = \frac{3}{2}$$

$$r = CA = |\alpha| = \frac{3}{2}$$

$$D = 3$$